

## Template for Computer Programming I Practical Recording

|                              |                                                                                                                                                                                                                                                                       |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Date</b>                  | : 27/03/2025                                                                                                                                                                                                                                                          |
| <b>Problem Specification</b> | : The program tracks sales data for multiple salespeople. It takes input for sales values, calculates the total and average sales, finds the highest and lowest sales, identifies the respective salespeople, and counts the number of sales exceeding a given limit. |
| <b>Assumption</b>            | :The number of salespeople (N) is provided as input.<br><br>Sales values are entered as positive integers.<br><br>There is at least one salesperson in the system.                                                                                                    |
| <b>Limitation</b>            | : The program does not handle negative sales values.<br><br>It assumes that all input values are valid integers.<br><br>If multiple salespeople have the same highest/lowest sales, the first occurrence is considered.                                               |
| <b>Input</b>                 | :Number of salespeople (N)<br><br>Sales amounts for each salesperson<br><br>Limit value for filtering sales above a threshold                                                                                                                                         |
| <b>Processing</b>            | : Accept sales input<br><br>Compute total and average sales<br><br>Determine the highest and lowest sales<br><br>Identify the salesperson with the highest and lowest sales<br><br>Count the number of sales exceeding the given limit                                |

**Output**

: List of sales amounts per salesperson  
Total sales amount  
Average sales amount  
Maximum and minimum sales values  
Salespeople with maximum and minimum sales  
Count of sales exceeding a given limit  
Salespeople whose exceeded the value

**Algorithm**

: Step1: Start  
Step2: Prompt the Enter the number of salespeoples for arraysize  
Step3: Initialize arraysize for user input  
Step4: Read the input given by the user  
Step5: Create an integer array sales[ ] of size n  
Step6: Prompt as Enter sales amount each of (arraysize) sales people  
Step7: Repeat i =0 to arraysize ,Read the input given by user for every sales[i] and store the sales amount for each salesperson in sales[i].  
  
Step8: Compute the total sum of all sales[I].  
  
Step9: Calculate the average sales value,  $avg \leftarrow \text{sum}/\text{arraysize}$   
  
Step10: Find the maximum sales and their salesperson ID  
Step11: Find the minimum sales and their salesperson ID  
Step12: Display the total, Average, Maximum, Minimum sales and their IDs  
Step13: Prompt the user to enter the value  
Step14: Initialize limitV for user input  
Step15: Read the user input  
Step16: Print salespeople IDs and sales amount of salespeople who exceed the value  
Step17: Initialize Count and display how many salespeople exceed the value  
  
Step18: End the program

**Programme listing** : Programme file **attached**

**Test data and expected output** : 1. Test data: 1.arraysize=5

2.sales =789  
452  
157  
456  
232

3. limitV=300

Expected output: 1 sales amount is 789  
2 sales amount is 452  
3 sales amount is 157  
4 sales amount is 456  
5 sales amount is 232

Sales total amount is : 2086  
Average Of Sales : 417.2

Maximum Of Sales : 789  
Salesperson 1 had the highest sale with \$789.

Minimum Of Sales : 157  
Salesperson 3 had the lowest sale with \$157.

The salespeople 1 whose sales amount is 789  
The salespeople 2 whose sales amount is 452  
The salespeople 4 whose sales amount is 456

The total number of salespeople whose sales  
exceeded the value=3

**Output obtained for test data** : 1. Test data: 1.arraysize=5  
2.sales  
789  
452  
157  
456  
232  
3. limitV=300

Expected output: 1 sales amount is 789  
2 sales amount is 452  
3 sales amount is 157  
4 sales amount is 456  
5 sales amount is 232

Sales total amount is : 2086  
Average Of Sales : 417.2

Maximum Of Sales : 789  
Salesperson 1 had the highest sale with \$789.

Minimum Of Sales : 157  
Salesperson 3 had the lowest sale with \$157.

The total number of salespeople whose sales exceeded the value=3

**Discussion** : The program successfully computes key sales metrics.

It accurately identifies the highest and lowest sales and respective salespeople.

The program performs well but does not handle invalid input

Enhancements can include input validation and better handling of ties for max/min sales.

## **Conclusion**

:The program effectively processes sales data, providing useful insights into total, average, highest, and lowest sales while identifying trends among salespeople